

6th semester

BRANCH	Subject Title	Subject Code
B.Sc.(Physics–DSE)	Astronomy and space Physics	PYDSE1-161

Course Learning Outcomes (CLO)

To study this course, the student must have had Physics as a subject in Diploma.

On completion of the course the student will be able to

1. Understand the structure of nucleus and nuclear energy
2. Understand the different forms of radioactivity and decay process
3. Develop the understanding for fission and fusion processes, Nuclear power Generation
4. Develop the understanding elementary particles and their properties

Unit:1

1. A Brief historical background of Astronomy and space physics, Introduction of five ancient astronomical observatories (Jantar Mantar) of India (Ujjain, Jaipur, Delhi, Mathur Varanasi), Contribution of Aryabhata, Brahmagupta, Bhaskara II, Vainu Bappu, Prof. Jayant Narlikar, Prof. Kasturi Rangan and Prof. S.K Mitra in the field of Astronomy and Space physics.

2. Astronomical Distances and their measurements, Determination of Mass. Concept of space and time, Luminosity. Temperature and distances of a stars. Stellar classification and its interpretation, H.R. diagram of clusters, Empirical mass- luminosity relation.

1. Positional astronomy. Celestial Sphere, Astronomical Coordinate Systems, Conversion of Coordinates.

Unit:2

Astronomical techniques

1. Basic Optical Definitions for Astronomy: Magnification, Light Gathering Power, Resolving Power, Diffraction Limit and Atmospheric Windows, Optical Telescopes: Types of Reflecting Telescopes, Telescope Mountings

2. Detectors and Their Use with Telescopes, Types of Detectors, Detection Limits with Telescopes,

Spectrograph, CCD Camera, Photometers, Filters. Polarimeter

2. Radio Telescope- Interferometry, UV. IR. X-ray and Gamma ray telescope. Space telescope Himalayan Chandra telescope (11ST) in India.

Unit:3

Evolution of Stars and their life cycle

1. Brightness, Radiant Flux and Luminosity, Apparent and Absolute magnitude scale, Distance Modulus, Determination of Temperature and Radius of a star, Stellar Interior, Energy generation in Stars, Contraction Hypothesis.
2. Evolution of Stars: Pre main sequence, Main sequence and post main sequence stages.
 1. Classification of stars, Binary, Neutron star, Black hole, Chandrashekhar limit.

Unit:4

Sun and Solar System

1. Solar Structure and its processes, Solar Atmosphere, Photosphere, Chromosphere, Corona, Concept of quiet and active Sun, Development of centre of activity, Sunspots, Butterfly diagram, Solar Cycle, Solar flares,
2. Solar System: Age, Planetary orbits and distance, Physical size, rotation periods.
3. Origin of the Solar System: The Nebular Model, Tidal Forces and Planetary Rings, Extra-Solar Planets

Unit:5

Galaxies

1. The Milky Way(our own galaxy): Basic Structure and Properties, Nature of Rotation, Stars and Star Clusters of the Milky Way, Dark Matter.
2. Morphology of galaxies, Classification of Galaxies, basic properties of Elliptical, Spiral and Seyfert Galaxies, Galactic clusters. Pulsars and Quasars.
 1. Gas and Dust in the Galaxy, Big Bang theory